Slide Set #3

Regression Models and the Classical Linear Regression Model (CLRM)

Pietro Coretto
pcoretto@unisa.it

Econometrics
Master in Economics and Finance (MEF)
Università degli Studi di Napoli "Federico II"

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Notes

Regression analysis

Regression analysis is a set of statistical techniques for modeling and analyzing the relationship between a *dependent* variable Y and one or more *independent variables* X. Typically $X = (X_1, X_2, \ldots, X_K)'$ is a vector of variables.

Depending on the context, and the field of application, we have different names

- Y: dependent variable, response variable, outcome variable, output variable, target variable etc.
- X: independent variable, regressor, covariate, explanatory variable, predictor, feature, etc.

In regression analysis we assume a certain mechanism linking the X to the Y. We want to use observed data to understand the link.

Notes

Regression models

Notes

Regression function

models the relationship between Y and X

$$Y \approx f(X)$$

in the sense that f approximates Y based on X

Conditional mean regression models

$$f(X) := \mathrm{E}[Y|X]$$

we are saying that we approximate ($\approx$) Y with its conditional expectation given the information generated by X

Notes

The quality of the approximation $Y \approx f(X)$ can be measure by the quadratic risk or MSE

$$\mathrm{E}[(Y - f(X))^2]$$

Theorem: Optimal Predictor

Under suitable regularity conditions: $\inf_f \mathrm{E}[(Y - f(X))^2] = \mathrm{E}[Y|X]$

In other words

- $f(X) = \mathrm{E}[Y|X]$ gives the best approximation to Y in terms of MSE
- If we want to guess Y based on information generated by X, $f(X) = \mathrm{E}[Y|X]$ is the best guess in terms of MSE

Proof. We need to show that for any function $f(X)$

$$\mathrm{E}[(Y - \mathrm{E}[Y|X])^2] \leq \mathrm{E}[(Y - f(X))^2]$$

$$\mathrm{E}[(Y - f(X))^2] = \mathrm{E}\left[\left(\underbrace{Y - \mathrm{E}[Y|X]}_{A} + \underbrace{\mathrm{E}[Y|X] - f(X)}_{B}\right)^2\right]$$

computing $(A+B)^2$ and using using the lineary of expectations

$$\mathrm{E}[A^2] + \mathrm{E}[B^2] + 2\,\mathrm{E}[AB] = \mathrm{E}[(Y - f(X))^2] \tag{3.1}$$

$\mathrm{E}[B^2] = \mathrm{E}[(\mathrm{E}[Y|X] - f(X))^2] \geq 0$, therefore (3.1) becomes

$$\mathrm{E}[A^2] + 2\,\mathrm{E}[AB] \;\leq\; \mathrm{E}[(Y - f(X))^2] \tag{3.2}$$

Notes

$$\begin{aligned}
\mathrm{E}[AB] &= \mathrm{E}[\,\mathrm{E}[AB\,|\,X]\,] \quad \text{(law of iterated expectations)} \\
&= \mathrm{E}\,[\,(Y - \mathrm{E}[Y|X])(\mathrm{E}[Y|X] - f(X)) \;|\; X] \\
&= (\mathrm{E}[Y|X] - f(X))\;\mathrm{E}\,[\,(Y - \mathrm{E}[Y|X]) \;|\; X] \quad \text{(pull out)} \\
&= (\mathrm{E}[Y|X] - f(X))\,\{\,\mathrm{E}[Y\,|\,X] - \mathrm{E}[\mathrm{E}[Y\,|\,X]\,|\,X]\} \\
&= (\mathrm{E}[Y|X] - f(X)) \times 0
\end{aligned}$$

Therefore, (3.2) becomes

$$\mathrm{E}[A^2] \;\leq\; \mathrm{E}[(Y - f(X))^2]$$

with $\mathrm{E}[A^2] = \mathrm{E}[(Y - \mathrm{E}[Y\,|\,X])^2]$, which proves the result ■

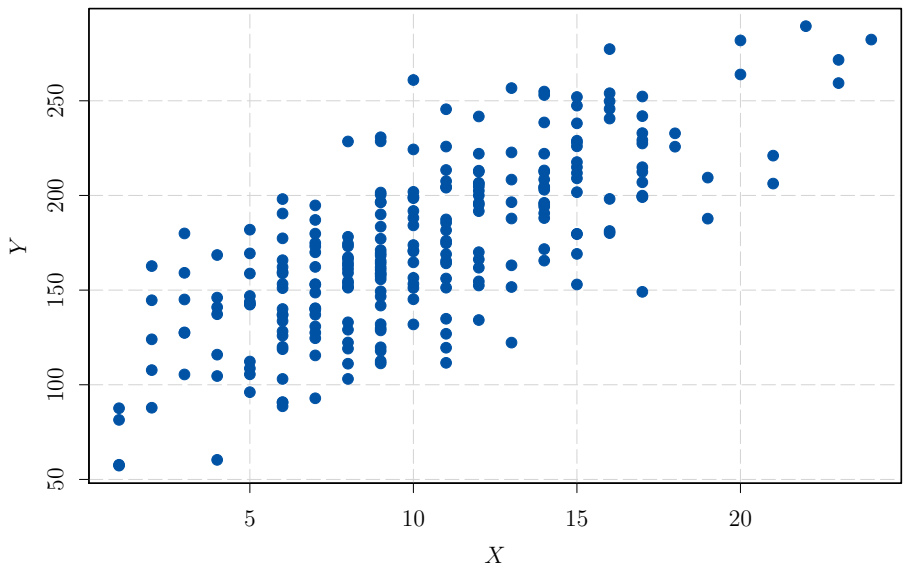
Notes

In this course we focus on conditional mean regression models where the regression function has a linear parametric form:

$$Y \approx \mathrm{E}[Y \mid X] = f(X; \boldsymbol{\beta}) = \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_K X_K$$

The reason why this class of regression models is so popular is that because they can reproduce correlation between Y and Xs. Going back to the example of slide set #1

Notes



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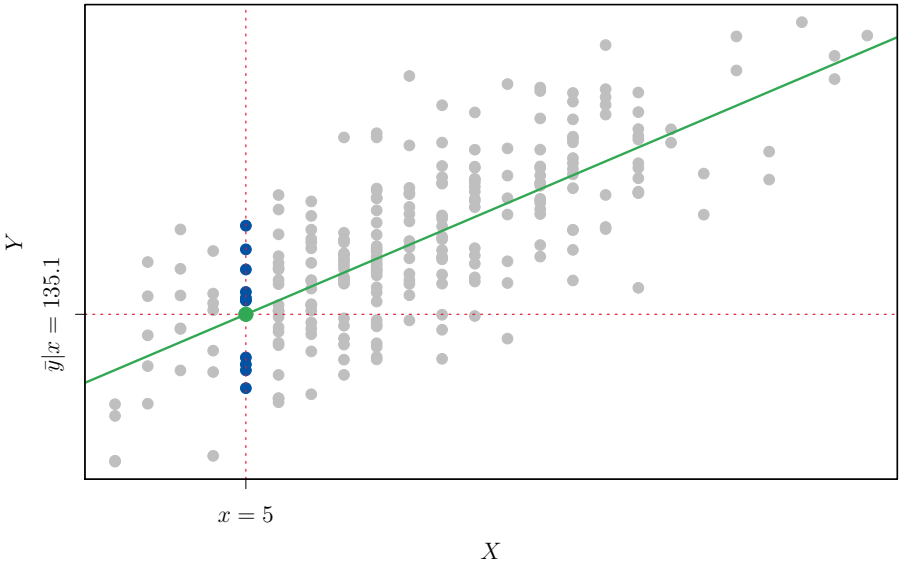

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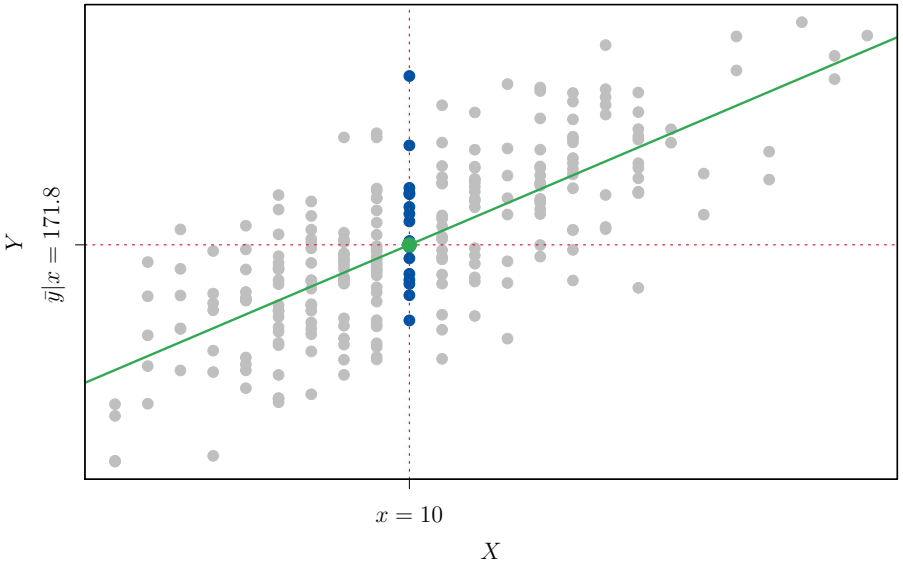

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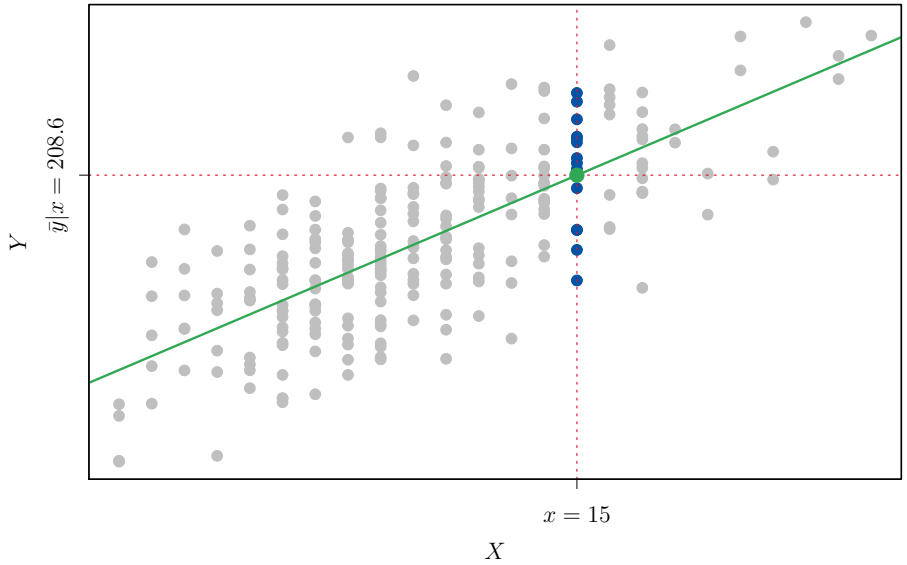

Notes

Sampling

in reality we cannot observe $\mathrm{E}[Y \mid X]$. For each sample unit $i = 1, 2, \ldots, n$ we observe

$$(Y_i, X_{i1}, X_{i2}, \ldots, X_{ik})$$

Linear mean regression model

a way to reproduce the previous mechanism is to add a random deviation (error term) from the mean

$$\begin{aligned} Y_i &= f(X_i; \boldsymbol{\beta}) + \varepsilon_i \\ &= \mathrm{E}[Y_i \mid X] + \varepsilon_i \\ &= \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_K X_{iK} + \varepsilon_i \end{aligned}$$

Notes

Partial or marginal effects

We want to measure of the effect on the regression function determined by a change in a regressor X_j holding all the other regressors constant (waee = "*with all else equal*").

Assuming differentiability of $f(\cdot)$, the *partial/marginal effect of a change* ΔX_j is given by

$$\Delta \, \mathrm{E}[Y \mid X] = \frac{\partial \, \mathrm{E}[Y \mid X]}{\partial X_j} \Delta X_j$$

hoding fixed all $X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_K$

Computing marginal/partial effects make sense only when the model has a *causal interpretation* (see later).

Notes

Linear regression with continuous X_j

$$\frac{\partial \, \mathrm{E}[Y \mid X]}{\partial X_j} = \beta_j$$

Therefore, the unknown parameter β_j coincides with *partial effect* of an *unit change* in X_j *waee*.

Linear regression with discrete X_j

In this case, partial effects are computed as the variations in $\mathrm{E}[Y \mid \boldsymbol{X}]$ obtained by changing the level of X_j and *waee*.

Suppose $X_k \in \{a, b, c\}$. The partial effect when X_k changes from level a to b (*waee*) is given by

$$\mathrm{E}[Y \mid X_k = b, \text{waee}] \; - \; \mathrm{E}[Y \mid X_k = a, \text{waee}]$$

Another measure of regressors' effect on the Y is the partial/marginal elasticity (see homeworks).

Notes

Notations

Indexes and constants:

- n: number of sample units
- K: number of covariates/regressors/features measured on each of the n sample units
- $i = 1, 2, \ldots, n$: indexes sample units
- $k = 1, 2, \ldots, K$: indexes regressors

$\boldsymbol{y} \in \mathbb{R}^n$: column vector of the dependent/response variable

$$\boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

Notes

$\boldsymbol{x}_i \in \mathbb{R}^K$: column vector containing the K regressors measured on the ith sample unit

$$\boldsymbol{x}_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iK} \end{pmatrix}$$

So called design matrix, is the $(n \times K)$ matrix whose rows contain sample units and columns contain regressors

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \ldots x_{1K} \\ x_{21} & x_{22} & \ldots x_{2K} \\ \vdots & \vdots & \ddots \vdots \\ x_{n1} & x_{n2} & \ldots x_{nK} \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1{}' \\ \boldsymbol{x}_2{}' \\ \vdots \\ \boldsymbol{x}_n{}' \end{pmatrix}$$

Notes

$\boldsymbol{\varepsilon} \in \mathbb{R}^n$: column vector containing the error term for each unit

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

Notes

Classical linear regression model (CLRM)

Assumption (A1): linearity

For all $i = 1, 2, ..., n$ observed data are generated by the following linear model

$$y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + ... + \beta_K x_{iK} + \varepsilon_i \quad (3.3)$$
$$= \boldsymbol{x}_i' \boldsymbol{\beta} + \varepsilon_i \quad (3.4)$$

where $\boldsymbol{\beta} \in \mathbb{R}^K$ is a vector of coefficients. In matrix form

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (3.5)$$

Remark 1: linearity of the model is wrt parameters not regressors. For x_2=log(consumption), the model is still linear wrt to log(consumption).

Notes

Remark 2: often a constant/intercept term is introduced in the model

$$y_i \;=\; \beta_1 + \beta_2 x_{i2} + ... + \beta_K x_{iK} + \varepsilon_i \quad \text{for } i = 1, 2, ..., n$$

In this case we consider conventionally (3.3) as if $x_{i1} = 1$ for all i in the sample.

If the model includes a constant/intercept term, then the first column of $\boldsymbol{X}$ is the unit column vector, that is $\boldsymbol{X}_{\cdot 1} = \mathbf{1}_n = (1, 1, \ldots, 1)'$

$$\boldsymbol{X} = \begin{pmatrix} 1 & x_{12} & \ldots x_{1K} \\ 1 & x_{22} & \ldots x_{2K} \\ \vdots & \vdots & \ddots \vdots \\ 1 & x_{n2} & \ldots x_{nK} \end{pmatrix}$$

Notes

Assumption (A2): strict exogeneity

$\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{X}] = 0$ for all $i = 1, 2, \ldots, n$, or $\mathrm{E}[\boldsymbol{\varepsilon} \,|\, \boldsymbol{X}] = \mathbf{0}$

Implications:

1. $\mathrm{E}[\varepsilon_i] = 0$ for all $i = 1, 2, \ldots, n$.
2. All regressors are orthogonal to the error term for all units: $\mathrm{E}[x_{jk}\varepsilon_i] = 0$ for all $i, j = 1, 2, \ldots, n$ and $k = 1, 2, \ldots, K$
3. Orthogonality implies the zero-correlation conditions $\mathrm{Cov}[x_{jk}, \varepsilon_i] = 0$ for all $i, j = 1, 2, \ldots, n$ and $k = 1, 2, \ldots, K$

⚠ If i =time (time series data), strict exogeneity implies that the error term is orthogonal to the past, current, and future regressors. For most time-series data, this condition is not satisfied, so the finite sample theory based on strict exogeneity is rarely applicable in time-series contexts.

Notes

Proof of the implications:

1. It follows from the law of iterated expectations:

$$\mathrm{E}[\varepsilon_i] = \mathrm{E}[\mathrm{E}[\varepsilon_i \,|\, \boldsymbol{X}]] = \mathrm{E}[0] = 0$$

2. By the law of iterated expectations write

$$\mathrm{E}[x_{ji}\varepsilon_i] = \mathrm{E}[\mathrm{E}[x_{jk}\varepsilon_i \,|\, x_{jk}]]$$

And by the linearity of the conditional expectation ("*pull out what's known*"), write

$$\mathrm{E}[\mathrm{E}[x_{jk}\varepsilon_i \,|\, x_{jk}]] = \mathrm{E}[x_{jk}E[\varepsilon_i \,|\, x_{jk}]]$$

But A2 says $E[\varepsilon_i \,|\, x_{jk}] = 0$, which proves the desired result.

3. This follows from previous results:

$$\mathrm{Cov}[x_{jk}, \varepsilon_i] = \mathrm{E}[x_{jk}\varepsilon_i] - E[x_{jk}]E[\varepsilon_i] = E[x_{jk}\varepsilon_i] = 0$$

Notes

Assumption (A3): absence of multicollinearity

The $(n \times K)$ design matrix $\boldsymbol{X}$ has rank K with probability 1.

This assumption implies that $\boldsymbol{X}$ has full column rank, which means that the columns of $\boldsymbol{X}$ are linearly independent

A3 also requires that $n \geq K$.

Essentially A3 is a technical assumption that guarantees that most computations can be performed... see this later.

Notes

Assumption A4: spherical error variance

For all $i, j = 1, 2, \ldots, n$ and $i \neq j$

1. $\mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] = \sigma^2 > 0$ (homoskedasticity)
2. $\mathrm{E}[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] = 0$ (units are uncorrelated)

A vector random variable is said to have a spherical distribution if its covariance matrix is a scalar multiple of the identity matrix. The *sphericity* here is shown as follows:

$$\mathrm{Var}[\varepsilon_i \mid \boldsymbol{X}] = \mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] - \mathrm{E}[\varepsilon_i \mid \boldsymbol{X}]^2 = \mathrm{E}[\varepsilon_i^2 \mid \boldsymbol{X}] = \sigma^2$$

and

$$\mathrm{Cov}[\varepsilon_i, \varepsilon_j \mid \boldsymbol{X}] = E[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] - E[\varepsilon_i \mid \boldsymbol{X}] E[\varepsilon_j \mid \boldsymbol{X}] = E[\varepsilon_i \varepsilon_j \mid \boldsymbol{X}] = 0$$

Now it is easy to show (do it as an exercise) that

$$\mathrm{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \boldsymbol{X}] = \mathrm{Var}[\boldsymbol{\varepsilon} \mid \boldsymbol{X}] = \sigma^2 \boldsymbol{I}_n$$

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